MSG No. 56.376 P_Iccn [Type IV, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `2'2'2`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[1,4,10,11]`
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[2,3,9,12]`
3	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[5,8,14,15]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[6,7,13,16]`

Table 2: Wyckoff site: `4b`, site symmetry: `2'2'2`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[1,4,10,11]`
2	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[2,3,9,12]`
3	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[5,8,14,15]`
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[6,7,13,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `..2/m'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[1,4,13,16]`
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[2,3,14,15]`
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[5,8,9,12]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[6,7,10,11]`

Table 4: Wyckoff site: `4d`, site symmetry: `..2/m'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[1,4,13,16]`
2	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[2,3,14,15]`
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[5,8,9,12]`
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[6,7,10,11]`

Table 5: Wyckoff site: `8e`, site symmetry: `-1`

No.	position	mapping
1	$[0, 0, 0]$	`[1,5]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,6]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,7]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,8]`
5	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[9,13]`
6	$\left[0, \frac{1}{2}, 0\right]$	`[10,14]`
7	$\left[\frac{1}{2}, 0, 0\right]$	`[11,15]`

8	$\left[0,\, 0,\, \frac{1}{2}\right]$	[12,16]

Table 6: Wyckoff site: 8f, site symmetry: 2’..

No.	position	mapping
1	$\left[x,\, \frac{3}{4},\, 0\right]$	[1,10]
2	$\left[x+\frac{1}{2},\, \frac{1}{4},\, \frac{1}{2}\right]$	[2,9]
3	$\left[-x,\, \frac{1}{4},\, \frac{1}{2}\right]$	[3,12]
4	$\left[\frac{1}{2}-x,\, \frac{3}{4},\, 0\right]$	[4,11]
5	$\left[-x,\, \frac{1}{4},\, 0\right]$	[5,14]
6	$\left[\frac{1}{2}-x,\, \frac{3}{4},\, \frac{1}{2}\right]$	[6,13]
7	$\left[x,\, \frac{3}{4},\, \frac{1}{2}\right]$	[7,16]
8	$\left[x+\frac{1}{2},\, \frac{1}{4},\, 0\right]$	[8,15]

Table 7: Wyckoff site: 8g, site symmetry: .2’.

No.	position	mapping
1	$\left[\frac{3}{4},\, y,\, 0\right]$	[1,11]
2	$\left[\frac{1}{4},\, -y,\, \frac{1}{2}\right]$	[2,12]
3	$\left[\frac{1}{4},\, y+\frac{1}{2},\, \frac{1}{2}\right]$	[3,9]
4	$\left[\frac{3}{4},\, \frac{1}{2}-y,\, 0\right]$	[4,10]
5	$\left[\frac{1}{4},\, -y,\, 0\right]$	[5,15]
6	$\left[\frac{3}{4},\, y,\, \frac{1}{2}\right]$	[6,16]
7	$\left[\frac{3}{4},\, \frac{1}{2}-y,\, \frac{1}{2}\right]$	[7,13]
8	$\left[\frac{1}{4},\, y+\frac{1}{2},\, 0\right]$	[8,14]

Table 8: Wyckoff site: 8h, site symmetry: ..2

No.	position	mapping
1	$\left[\frac{3}{4},\, \frac{3}{4},\, z\right]$	[1,4]
2	$\left[\frac{1}{4},\, \frac{1}{4},\, \frac{1}{2}-z\right]$	[2,3]
3	$\left[\frac{1}{4},\, \frac{1}{4},\, -z\right]$	[5,8]
4	$\left[\frac{3}{4},\, \frac{3}{4},\, z+\frac{1}{2}\right]$	[6,7]
5	$\left[\frac{1}{4},\, \frac{1}{4},\, z+\frac{1}{2}\right]$	[9,12]
6	$\left[\frac{3}{4},\, \frac{3}{4},\, -z\right]$	[10,11]
7	$\left[\frac{3}{4},\, \frac{3}{4},\, \frac{1}{2}-z\right]$	[13,16]
8	$\left[\frac{1}{4},\, \frac{1}{4},\, z\right]$	[14,15]

Table 9: Wyckoff site: 8i, site symmetry: ..2

No.	position	mapping
1	$\left[\frac{3}{4},\, \frac{1}{4},\, z\right]$	[1,4]
2	$\left[\frac{1}{4},\, \frac{3}{4},\, \frac{1}{2}-z\right]$	[2,3]
3	$\left[\frac{1}{4},\, \frac{3}{4},\, -z\right]$	[5,8]

continued ...

Table 9

No.	position	mapping
4	$\left[\frac{3}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	[6,7]
5	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	[9,12]
6	$\left[\frac{3}{4}, \frac{1}{4}, -z\right]$	[10,11]
7	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	[13,16]
8	$\left[\frac{1}{4}, \frac{3}{4}, z\right]$	[14,15]

Table 10: Wyckoff site: 8j, site symmetry: ..m'

No.	position	mapping
1	$\left[x, y, \frac{3}{4}\right]$	[1,16]
2	$\left[x+\frac{1}{2}, -y, \frac{3}{4}\right]$	[2,15]
3	$\left[-x, y+\frac{1}{2}, \frac{3}{4}\right]$	[3,14]
4	$\left[\frac{1}{2}-x, \frac{1}{2}-y, \frac{3}{4}\right]$	[4,13]
5	$\left[-x, -y, \frac{1}{4}\right]$	[5,12]
6	$\left[\frac{1}{2}-x, y, \frac{1}{4}\right]$	[6,11]
7	$\left[x, \frac{1}{2}-y, \frac{1}{4}\right]$	[7,10]
8	$\left[x+\frac{1}{2}, y+\frac{1}{2}, \frac{1}{4}\right]$	[8,9]

Table 11: Wyckoff site: 16k, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$\left[x+\frac{1}{2}, -y, \frac{1}{2}-z\right]$	[2]
3	$\left[-x, y+\frac{1}{2}, \frac{1}{2}-z\right]$	[3]
4	$\left[\frac{1}{2}-x, \frac{1}{2}-y, z\right]$	[4]
5	$[-x, -y, -z]$	[5]
6	$\left[\frac{1}{2}-x, y, z+\frac{1}{2}\right]$	[6]
7	$\left[x, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[7]
8	$\left[x+\frac{1}{2}, y+\frac{1}{2}, -z\right]$	[8]
9	$\left[x+\frac{1}{2}, y+\frac{1}{2}, z+\frac{1}{2}\right]$	[9]
10	$\left[x, \frac{1}{2}-y, -z\right]$	[10]
11	$\left[\frac{1}{2}-x, y, -z\right]$	[11]
12	$\left[-x, -y, z+\frac{1}{2}\right]$	[12]
13	$\left[\frac{1}{2}-x, \frac{1}{2}-y, \frac{1}{2}-z\right]$	[13]
14	$\left[-x, y+\frac{1}{2}, z\right]$	[14]
15	$\left[x+\frac{1}{2}, -y, z\right]$	[15]
16	$\left[x, y, \frac{1}{2}-z\right]$	[16]